Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/07/2024
PI Name:	Temple, David Ph.D.		
Project Title:	Systematic Assessment of Noisy Galvanic Vestibular Stimulation as a Sensorimotor Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80303-7813	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 TRISH-RFA-2201-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Torin Ph.D. (MENTOR: University of Colorado, Boulder)		
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Task Description:	POSTDOCTORAL FELLOWSHIP Here, we propose noisy Galvanic Vestibular Stimulation (nGVS) as a potential countermeasure to mitigate sensorimotor disruptions experienced by astronauts during gravity transitions. nGVS is thought to improve sensory information throughput via a phenomenon known as stochastic resonance (SR). Unfortunately, several gaps in the literature exist that prevent the utilization of nGVS as a viable countermeasure for astronauts. We aim to systematically assess the efficacy of utilizing nGVS as a countermeasure to improve astronaut performance through: 1) optimizing the waveform of the noisy electrical current for vestibular enhancement, 2) determining if optimal vestibular enhancement generalizes to optimal performance on a number of balance, locomotor, and manual control tasks, 3) investigating potential after-effects of the nGVS on performance, and 4) testing the effects of nGVS in a gravity transition analog, with the goal of increasing the countermeasure readiness level. We propose to accomplish these objectives through a series of human subject experiments systematically testing various nGVS waveforms' impact on vestibular perceptual thresholds to determine the optimal waveform for inducing SR within the vestibular system (Aim 1). Then we will implement the optimal vestibular perceptual nGVS waveform in various balance, locomotor, and manual control nulling tasks to determine if relationships exist between SR-induced optimal vestibular performance and optimal performance in various operational tasks performed by astronauts (Aim 2). Next, we will assess vestibular perceptual thresholds and balance performance metrics prior to, during, immediately after, and 1 hour after prolonged exposure to nGVS to evaluate any potential aftereffects nGVS might have on the functional performance (Aim 3). Finally, we will use a ground-based analog for gravity transitions relevant for astronauts, which applies sustained hyper-gravity centrifugation (Aim 4). This analog induces sensorimotor disturbances following centrifugation, similar to post-flight astronauts, and we will test whether the optimized nGVS application can mitigate balance impairment. If successful, this research effort will deliver a validated countermeasure for astronaut sensorimotor impairment due to gravity transitions. Critically, nGVS is a low-cost, mass, power, and volume approach that is safe, well-tolerated, and could be utilized during long-duration exploration in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We proposed noisy Galvanic Vestibular Stimulation (nGVS) as a potential countermeasure to mitigate sensorimotor disruptions experienced by astronauts during gravity transitions. nGVS is thought to improve sensory information throughput via a phenomenon known as stochastic resonance (SR). Unfortunately, several gaps in the literature exist that prevent utilization of nGVS as a viable countermeasure for astronauts. We aimed to systematically assess the efficacy of utilizing nGVS as a countermeasure to improve astronaut performance through: 1) optimizing the waveform of the noisy electrical current for vestibular enhancement, 2) determining if optimal vestibular enhancement generalizes to optimal performance on a number of balance, locomotor, and manual control tasks, 3) investigating potential after-effects of the nGVS on performance, and 4) testing the effects of nGVS in a gravity transition analog, with the goal of increasing the countermeasure readiness level. We proposed to accomplish these objectives through a series of human subject experiments systematically testing various nGVS waveforms' impact on vestibular perceptual thresholds to determine the optimal waveform for inducing SR within the vestibular system (Aim 1). Then we will implement the optimal vestibular perceptual nGVS waveform in various balance, locomotor, and manual control nulling tasks to determine if relationships exist between SR-induced optimal vestibular performance and optimal performance in various operational tasks performed by astronauts (Aim 2). Next, we will assess vestibular perceptual thresholds and balance performance metrics prior to, during, immediately after, and 1 hour after prolonged exposure to nGVS to evaluate any potential aftereffects nGVS might have on the functional performance (Aim 3). Finally, we will use a ground-based analog for gravity transitions relevant for astronauts, that applies sustained hyper-gravity centrifugation (Aim 4). This analog induces sensorimotor disturbances following centrifugation, similar to post-flight astronauts, and we will test whether the optimized nGVS application can mitigate balance impairment. If successful, this research effort will deliver a validated countermeasure for astronaut sensorimotor impairment due to gravity transitions. Critically, nGVS is a low cost, mass, power, and volume approach that is safe, well-tolerated, and could be utilized during long-duration, exploration space missions. We anticipate it may enable critical mission functions immediately upon crewmembers landing on the Moon or Mars. Upon beginning vestibular perceptual threshold testing in Aim 1, it became evident that subject fatigue from performing hundreds of translation trials could negatively impact computed vestibular perceptual thresholds and likewise the opportunity to see performance curves indicative of SR-like phenomena at various nGVS amplitudes. Thus, we developed a new aim (Aim 0) to determine if utilizing a Confidence Signal Detection (CSD) model, where subjects report confidence in their left/right responses (from 50-100%), could drastically reduce the number of trials needed to compute accurate vestibular perceptual thresholds. To be successful, reduced numbers of vestibular perceptual threshold trials per nGVS amplitude must: 1) compute threshold values similar to those performed with the gold standard binary thresholding methods (performing 100 trials per nGVS amplitude), and 2) reproduce similar SR-like phenomena as observed with traditional binary thresholding methods. Currently we have collected CSD data in 12 subjects and are finalizing our analysis to determine how many vestibular perceptual threshold trials can be reduced at each nGVS level to limit the impact of subject fatigue. In regard to Aim 1, a 0-30 Hz nGVS profile has yielded little evidence of SR occurrence in four subjects that were tested. It should be noted, however, that this 0-30 Hz nGVS signal is not the same one as has been previously tested in 2018 by Galvan-Garza et al. We have tested eight subjects with a 0-2 Hz nGVS profile that appears to show potential SR occurrence in vestibular perceptual thresholds for some of the subjects. We are currently updating parameters for SR curves and expect classification of SR occurrence to be finalized in these two tested nGVS profiles by the completion of year one. In the coming year, we expect to proceed with Aims 2-4 with fewer trials being performed than initially proposed during vestibular perceptual threshold tests, by utilizing a CSD model. References (Ed., 1/7/24) Galvan-Garza, R. C., Clark, T. K., Mulavara, A. P., and Oman, C. M. (2018). Exhibition of stochastic resonance in vestibular tilt motion perception. Brain Stimul. 11, 716–722. doi: 10.1016/j.brs.2018.03.017

Bibliography Type:	Description: (Last Updated: 03/12/2025)
Abstracts for Journals and Proceedings	Temple DR. "Potential spaceflight applications for galvanic vestibular stimulation." International Space Development Conference, Frisco, TX, May 25-28, 2023. Abstracts. International Space Development Conference, Frisco, TX, May 25-28, 2023. , May-2023
Abstracts for Journals and Proceedings	Temple DR, Clark TK. "Assessing noisy galvanic vestibular stimulation capabilities for inducing stochastic resonance in lateral translation vestibular perceptual thresholds." Vestibular-Oriented Research Meeting, Boulder, CO, June 25-29, 2023. Abstracts. Vestibular-Oriented Research Meeting, Boulder, CO, June 25-29, 2023. , Jun-2023
Abstracts for Journals and Proceedings	Temple DR, Clark TK. "Systematic assessment of noisy galvanic vestibular stimulation as a sensorimotor countermeasure." 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, "To the Moon: The Next Golden Age of Human Spaceflight", Galveston, TX, February 7-9, 2023. , Feb-2023
Awards	Temple D. "2nd Place in the 2023 NASA Human Research Program Investigators' Workshop Post Doctoral Fellows Poster Competition, February 2023." Feb-2024
Awards	Temple D. "College of Engineering & Applied Science "Outstanding Mentor Award", October 2022." Oct-2022